

Worker KPI Dashboard

End User Guide

Camtronics MES Module

Version 18.0

Production Worker Performance Tracking System

Table of Contents

1. Introduction	3	
2. Accessing the Dashboard		4
3. Understanding KPI Metrics		5
4. Creating a KPI Report		7
5. Reading the Dashboard		9
6. Printing Reports	11	
7. Best Practices	12	
8. Frequently Asked Questions		13

1. Introduction

The Worker KPI Dashboard is a powerful tool designed to help production managers and supervisors track and analyze the performance of production workers. This dashboard provides key metrics that help identify top performers, areas for improvement, and overall production efficiency.

Purpose

The Worker KPI Dashboard helps you:

- Monitor individual worker productivity and efficiency
- Track idle time to optimize workforce utilization
- Identify quality issues through reject rate analysis
- Compare worker performance over different time periods
- Generate printable reports for management review

Key Features

Real-time Calculations: KPIs are calculated based on actual completed work orders

Flexible Periods: Analyze daily, weekly, monthly, or custom date ranges

Worker Filtering: View all workers or focus on a specific individual

Visual Indicators: Color-coded status helps quickly identify performance levels

PDF Reports: Generate professional reports for printing or sharing

2. Accessing the Dashboard

Navigation Path

To access the Worker KPI Dashboard, follow these steps:

Step 1: Log in to the system with your credentials

Step 2: Click on the "MES" menu in the main navigation bar

Step 3: Select "Planning" from the dropdown menu

Step 4: Click on "Worker KPI Dashboard"

Menu Path: MES → Planning → Worker KPI Dashboard

Required Permissions

Access to the Worker KPI Dashboard depends on your user role:

Role	View	Create	Edit	Delete
MES User	Yes	No	No	No
MES Planner	Yes	Yes	Yes	No
MES Manager	Yes	Yes	Yes	Yes

Contact your system administrator if you need additional access permissions.

3. Understanding KPI Metrics

The Worker KPI Dashboard tracks six key performance indicators. All metrics are calculated based on COMPLETED work orders only, ensuring accurate performance measurement.

3.1 Idle Time Percentage

What it measures: The percentage of scheduled working time when the worker was not actively working on production tasks.

Formula:

$$(Total\ Working\ Hours - Total\ Real\ Duration) \div Total\ Working\ Hours \times 100\%$$

Interpretation: Lower is better. A high idle time percentage may indicate scheduling issues, material shortages, or equipment problems.

Idle Time %	Status	Meaning
0% - 20%	Good	Worker is well-utilized
21% - 40%	Warning	Some idle time, review causes
Above 40%	Critical	Significant idle time, action needed

3.2 Productivity (PCS/HR)

What it measures: The number of pieces (units) produced per working hour.

Formula:

$$Total\ Quantity\ Produced \div Total\ Working\ Hours$$

Interpretation: Higher is better. This metric helps compare output rates across workers, considering the time they spent on the production floor.

3.3 Production Efficiency Percentage

What it measures: How actual production time compares to the planned/expected time.

Formula:

$$Total\ Expected\ Duration \div Total\ Real\ Duration \times 100\%$$

Interpretation:

Efficiency %	Meaning
100%	Work completed exactly as planned
Above 100%	Work completed faster than planned (excellent)
Below 100%	Work took longer than planned

3.4 Reject Rate Percentage

What it measures: The percentage of produced items that were rejected or scrapped due to quality issues.

Formula:

$$\text{Rejected Quantity} \div \text{Total Quantity Produced} \times 100\%$$

Interpretation: Lower is better. A high reject rate indicates quality issues that may require additional training or process improvements.

3.5 Completed Work Orders

What it measures: The total count of work orders completed by the worker during the selected period.

Interpretation: This number should be considered alongside other metrics. A high count with low efficiency may indicate rushing, while a low count with high efficiency may indicate complex work assignments.

3.6 Quantity Processed

What it measures: The total quantity of products processed across all completed work orders.

Interpretation: This is the sum of all quantities produced. It provides a raw output number without time normalization.

4. Creating a KPI Report

Step-by-Step Instructions

Step 1: Open the Dashboard

Navigate to MES → Planning → Worker KPI Dashboard and click the "Create" button.

Step 2: Select the Period

Choose your reporting period from the dropdown options:

Period Type	Description	Date Range
Daily	Single day analysis	Today only
Weekly	Current week	Monday to today
Monthly	Current month	1st of month to today
Custom	User-defined range	Any dates you specify

The date fields will automatically adjust based on your selection. For custom periods, manually enter the start and end dates.

Step 3: Select Worker (Optional)

You can either:

- Leave the Worker field empty to analyze ALL production workers
- Select a specific worker to view only their individual performance

Step 4: Calculate KPIs

Click the **"Calculate KPIs" button** at the top of the form. The system will:

- Find all completed work orders in the selected period
- Match work orders to workers through daily assignments
- Retrieve attendance records for working hours
- Calculate all KPI metrics for each worker
- Display results in the Worker Details tab

Step 5: Review Results

The dashboard will display:

- Summary cards at the top showing averages across all workers
- Detailed worker-by-worker breakdown in the table below
- Color-coded rows indicating performance levels

5. Reading the Dashboard

5.1 Summary Cards

At the top of the dashboard, you will see six summary cards that provide a quick overview of all workers combined:

Card	Shows	Color
Workers	Total number of workers analyzed	Blue
Avg Idle Time	Average idle time percentage	Yellow
Avg Productivity	Average pieces per hour	Green
Avg Efficiency	Average efficiency percentage	Light Blue
Avg Reject Rate	Average reject rate percentage	Red
Total Qty	Total quantity processed + WO count	Gray

5.2 Worker Details Table

The main table shows individual performance for each worker. The columns include:

Column	Description
Worker	Employee name
Department	Worker's department
Idle Time %	Progress bar showing idle time (lower is better)
Productivity	Pieces produced per hour
Efficiency %	Progress bar showing efficiency (higher is better)
Reject %	Percentage of rejected products
Completed WO	Number of work orders completed
Qty Processed	Total quantity produced

5.3 Color-Coded Performance Indicators

Each row in the worker details table is color-coded based on efficiency performance:

Row Color	Efficiency Level	Action
Green	90% and above	Excellent - No action needed
Orange/Yellow	70% to 89%	Warning - Review and monitor
Red	Below 70%	Critical - Investigate immediately

5.4 Optional Columns

Some columns are hidden by default but can be displayed. Click on the column selector (three dots) at the right side of the table header to show/hide:

- Employee Badge/Barcode
- Qty Rejected (actual rejected quantity)
- Total Working Hours
- Total Real Duration
- Total Expected Duration

5.5 Formulas Tab

Click on the "Formulas" tab to view a reference guide explaining how each KPI is calculated. This is helpful when explaining the metrics to team members or during performance reviews.

6. Printing Reports

Generating a PDF Report

1. Ensure you have calculated the KPIs first
2. Click the "Print Report" button at the top of the form
3. A PDF document will be generated and downloaded
4. Open the PDF to view, print, or share with colleagues

Report Contents

The printed report includes:

- Report title and date range
- Summary table with averages for all workers
- Detailed table showing each worker's KPIs
- Color highlighting for performance levels
- Formula reference section at the bottom
- Generation timestamp

Tips for Printing

- Use landscape orientation for better table visibility
- Print in color to preserve performance indicators
- Save the PDF for record-keeping before printing

7. Best Practices

7.1 Regular Monitoring

- Generate weekly KPI reports for routine monitoring
- Use monthly reports for performance reviews
- Compare trends across multiple periods to identify patterns

7.2 Data Quality

For accurate KPIs, ensure:

- Attendance records are updated daily
- Work orders are marked as complete when finished
- Rejected quantities are recorded on work orders
- Workers are properly assigned through daily schedules

7.3 Using KPIs Constructively

- Focus on improvement trends, not just absolute numbers
- Consider external factors (machine breakdowns, material issues)
- Use KPIs as coaching tools, not punitive measures
- Recognize and reward high performers
- Investigate root causes of low performance before drawing conclusions

7.4 Comparing Workers Fairly

When comparing workers, consider that different work orders have different complexity levels. A worker assigned to complex assemblies may have lower productivity numbers but higher quality. Always review multiple metrics together rather than focusing on a single KPI.

8. Frequently Asked Questions

Q: Why does a worker show all zeros?

A: The worker may not have any completed work orders in the selected period, or they may not be linked to work orders through daily assignments. Check that work orders are marked as "Done" and that the worker was properly assigned.

Q: Why is the Idle Time showing 100%?

A: This means the worker has scheduled working hours (attendance record) but no recorded production time from completed work orders. Either no work orders were completed, or the worker was not assigned to any work orders during this period.

Q: Why is Efficiency over 100%?

A: Efficiency above 100% means the worker completed work faster than the expected/planned time. This is a positive indicator showing the worker exceeded expectations.

Q: Can I see KPIs for past months?

A: Yes! Use the Custom period type and set the date range to any historical period. Previous KPI records are also saved and can be accessed from the list view.

Q: Why are some workers not appearing?

A: Only employees marked as "Production Worker" in their employee profile will appear in the Worker KPI Dashboard. Contact your administrator to update employee settings.

Q: How often should I calculate KPIs?

A: This depends on your needs. For daily operations monitoring, calculate daily. For performance reviews, weekly or monthly calculations are recommended.

Q: Can I export the data to Excel?

A: Yes! In the Worker Details tab, click the export icon (download arrow) to export the table data to Excel format for further analysis.

Q: What if attendance records are missing?

A: If attendance records are not available, working hours will show as zero, which affects Idle Time and Productivity calculations. Ensure attendance is recorded daily for accurate KPIs.